How the two lines of equations 15 and 16 connect

Clarification of the urban compact methods report

The individual flows enter through a **total-allocation constraint**: lane additions must sum to the admitted mass, and leg discharge rates must sum to the usable output rate. Substituting the lower cases into the derivative of that constraint produces the upper equation.

Equation 15: lane additions share one mass budget. Here x_ℓ is the mass of the current packet actually admitted to lane ℓ. It determines how much traffic enters that lane; it is not a discharge rate. The total admitted mass is V. The starting load is h_ℓ, the admission bound is u_ℓ, and λ is the common filling level.

On an unchanged set of allocation categories, the forward rule is

$$x_\ell = \begin{cases} \lambda - h_\ell, & \ell \in \mathcal{A} \quad (0 < x_\ell < u_\ell), \\ u_\ell, & \ell \in \mathcal{U} \quad \text{(upper bound binds)}, \\ 0, & \text{lane remains without an addition}, \end{cases} \qquad \sum_\ell x_\ell = V.$$

Thus x_ℓ matters even though it does not appear explicitly in the upper gradient formula: its forward value identifies which case applies, and its derivative must respect the total mass budget.

Differentiate the total, then replace each individual derivative by the appropriate lower case of equation 15:

$$\underbrace{\frac{\mathrm{d}V}{\mathrm{d}\alpha} = \sum_\ell \frac{\mathrm{d}x_\ell}{\mathrm{d}\alpha}}_{\text{differentiated total-mass constraint}} \qquad = \sum_{\ell\in\mathcal{A}} \left(\frac{\mathrm{d}\lambda}{\mathrm{d}\alpha} - \frac{\mathrm{d}h_\ell}{\mathrm{d}\alpha} \right) + \sum_{\ell\in\mathcal{U}} \frac{\mathrm{d}u_\ell}{\mathrm{d}\alpha}.$$

Lanes remaining at zero contribute zero. Each lane in $\mathcal{A}$ contains the same unknown $\mathrm{d}\lambda/\mathrm{d}\alpha$, so collecting those terms gives

$$|\mathcal{A}| \frac{\mathrm{d}\lambda}{\mathrm{d}\alpha} = \frac{\mathrm{d}V}{\mathrm{d}\alpha} - \sum_{\ell\in\mathcal{U}} \frac{\mathrm{d}u_\ell}{\mathrm{d}\alpha} + \sum_{\ell\in\mathcal{A}} \frac{\mathrm{d}h_\ell}{\mathrm{d}\alpha}.$$

Dividing by $|\mathcal{A}|$ gives the upper line of equation 15. **You substitute the lower cases into the total constraint, rather than substituting the forward values of x_ℓ into the upper line.**

For calculation, first solve the forward allocation and identify its categories. Given the input sensitivities of V, u_ℓ and h_ℓ, use the upper line to obtain the filling-level sensitivity. Then use the lower line to obtain each lane's admitted-mass sensitivity. Their sum equals the sensitivity of V by construction.

Equation 16: inbound legs share one output-rate budget. Here r_{ij} is the discharge rate granted to inbound leg i toward output j. The lane-sharing calculation divides this grant among that leg's ready lanes; their rates determine transferred mass over each head-service interval. The total grant toward j is W_j.

For a fixed output j, the cap of leg i is C_{ij}, its green weight is w_{ij}, and θ_j is the common multiplier of those weights. With $\mathcal{C}$ the capped legs and $\mathcal{F}$ the uncapped legs,

$$r_{ij} = \begin{cases} C_{ij}, & i \in \mathcal{C}, \\ \theta_j w_{ij}, & i \in \mathcal{F}, \end{cases} \qquad \sum_i r_{ij} = W_j.$$

Again, differentiate the total constraint and substitute the lower cases of equation 16. The uncapped case uses the product rule:

$$\begin{aligned} \frac{\mathrm{d}W_j}{\mathrm{d}\alpha} &= \sum_i \frac{\mathrm{d}r_{ij}}{\mathrm{d}\alpha} \\ &= \sum_{i \in \mathcal{C}} \frac{\mathrm{d}C_{ij}}{\mathrm{d}\alpha} + \sum_{i \in \mathcal{F}} \left(w_{ij} \frac{\mathrm{d}\theta_j}{\mathrm{d}\alpha} + \theta_j \frac{\mathrm{d}w_{ij}}{\mathrm{d}\alpha} \right). \end{aligned}$$

Collect the terms containing the same multiplier sensitivity:

$$\left(\sum_{i \in \mathcal{F}} w_{ij} \right) \frac{\mathrm{d}\theta_j}{\mathrm{d}\alpha} = \frac{\mathrm{d}W_j}{\mathrm{d}\alpha} - \sum_{i \in \mathcal{C}} \frac{\mathrm{d}C_{ij}}{\mathrm{d}\alpha} - \theta_j \sum_{i \in \mathcal{F}} \frac{\mathrm{d}w_{ij}}{\mathrm{d}\alpha}.$$

Dividing by the total uncapped weight gives the upper line of equation 16. The individual rates disappear explicitly because their lower-case expressions have replaced them inside their sum.

The calculation again runs from the common quantity to the individual ones: solve the forward rates and identify the capped legs; use input sensitivities of total service, caps and weights to calculate the upper line; then calculate each leg's rate sensitivity using the lower line.

Why the two lines are not circular. In the derivation, individual responses are inserted into a shared budget to solve for one unknown common-level response. In numerical evaluation, that common-level response is calculated first and then distributed back to the individual flows. Neither upper line requires the individual flow derivatives as an additional unknown input.

These formulas apply while the categories remain unchanged and there is at least one uncapped contributor (with positive total weight in equation 16). If none remains, capped flows follow their caps directly; there is no need to determine a common-level derivative by division.